

Functions

Learning Objectives

- Use SUM, AVERAGE, COUNT, MAX, and MIN functions
- Enter functions using the function wizard
- Apply functions to real-world data sets

Engage (5-7 minutes)

Think of a teacher who needs to calculate the average marks of 40 students. Writing $=A1+A2+\dots+A40$ is tedious. Functions like SUM and AVERAGE do this in one step—they are shortcuts for common calculations that save hours of work.

Explore (10-12 minutes)

Enter marks for 5 subjects in A1:A5 (85, 90, 78, 92, 88). In B1, type $=\text{SUM}(A1:A5)$ to find total. In B2, type $=\text{AVERAGE}(A1:A5)$ for average. In B3, type $=\text{COUNT}(A1:A5)$ to count entries. In B4, type $=\text{MAX}(A1:A5)$ for highest. In B5, type $=\text{MIN}(A1:A5)$ for lowest.

Explain (10-12 minutes)

Functions are built-in formulas for common calculations. SUM adds a range, AVERAGE finds the mean, COUNT counts numbers, MAX finds the largest, MIN finds the smallest. Use colon (:) to specify ranges like A1:A5. The function wizard (fx button) helps build complex functions step by step.

Elaborate (8-10 minutes)

Create a farmer's crop yield sheet. Enter yields for 6 months in B1:B6. Use SUM to find total annual yield, AVERAGE for monthly average, MAX for best month, MIN for worst month. Add a COUNT to verify all months are recorded. Format results with appropriate decimal places.

Evaluate (5-8 minutes)

Answer: What is the difference between SUM and AVERAGE? When would you use COUNT instead of SUM? Write a function to find the range (MAX minus MIN) of temperatures in C1:C7.